

# SMART WHEELCHAIR: AUTONOMOUS PATIENT NAVIGATION IN HOSPITAL ENVIRONMENTS

AI-Driven intelligent navigation

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# Real-World Problem & Possible Approaches

## The Challenge of Hospital Transport Among Hospitalized Patients:

- Hospitals rely on staff to **manually** escort patients between departments
- In large hospitals, distances between departments are significant
- Emergency cases require fast, reliable routing – **delays have consequences**

## Project's Approach:

- Can an autonomous wheelchair navigate optimally through a hospital interior, carrying a patient?

### Attacking The Problem:

- For navigation, the continuous floor space must be divided into small blocks
- The project must simulate non-static, dynamic behaviors and environment, for more realistic simulation

# Implemented Solution

## Tools:

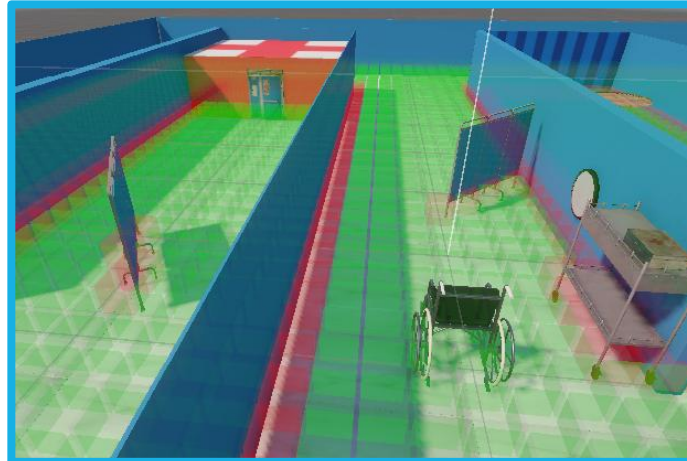
- Unity Engine (Physics & Visualization)
- C# Programming Language (Controller Scripts)
- Royalty free 3D assets found on the internet

## Navigation:

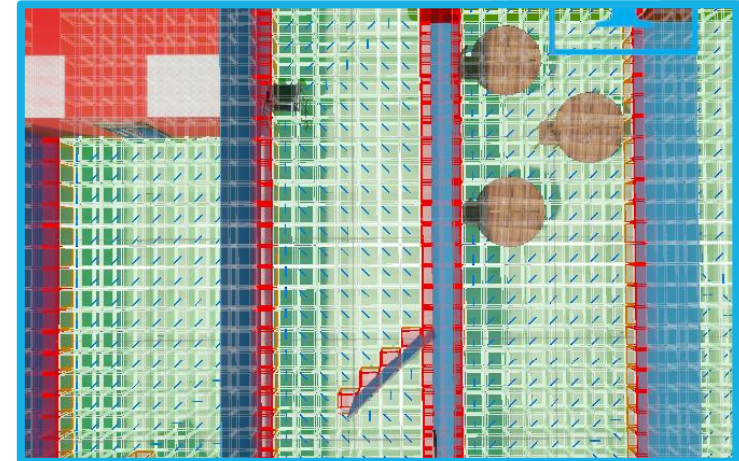
For the agent's navigation, the Flow Field pathfinding algorithm was implemented. How? The hospital floor is divided into small cells of variable size (based on user's preferences). Each cell is assigned a directional vector pointing toward the lowest-cost path to the set destination, allowing the wheelchair to navigate simply by reading the direction of whichever cell it currently occupies.



*(Grid & cells before runtime)*



*(Unwalkable mask on grid load)*



*(Cells with flow field vectors on runtime)*

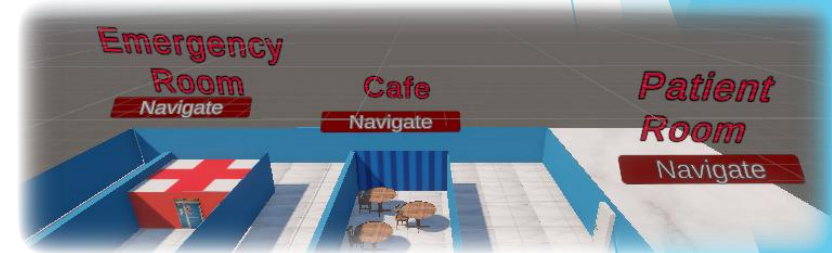
# Implemented Solution

## UI Control VS Simulation:

How is the agent's destination specified (from 3 specific destinations available)?

### 1. UI Navigation Buttons

- Over each destination, a UI Button “Navigate” appears.
- On click, the agent sets its destination to the respective location
- The flow field agent controller moves the wheelchair accordingly



### 2. Simulation loop

- The UI has 2D buttons “Simulation” and “Stop Simulation”.
- On simulation clicked, the HospitalSimulator controller picks a location randomly, waits for the agent to arrive and loops again, simulating the patient needs
- Each location pick percentage is changed dynamically and cannot be picked twice in a row
- UI Text describes the situation of the simulation accordingly, accompanied by sim. counter





# Challenges Encountered During Development

## Agent Getting Stuck on Unwalkable Cells

There was a small chance that the wheelchair would freeze when the flow field returned a zero vector near obstacle edges — resolved by implementing a stuck detection timer and a forward force mechanism to recover the agent automatically.

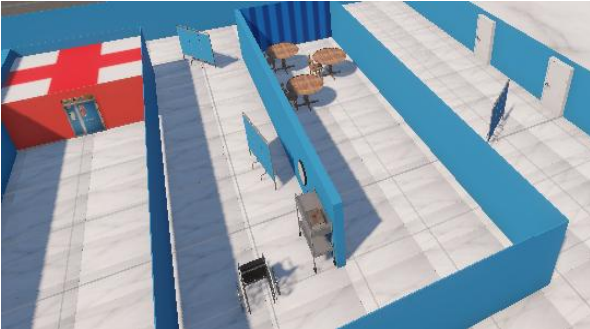
## Character Size vs Grid Resolution

The grid's cells had inappropriate size relative to the agent's collision sphere. As a result, the unwalkable mask was wrong. Therefore, either the agent would approach too tightly the obstacles, or the agent would steer too far from an obstacle. The grid & cells size was fine-tuned.

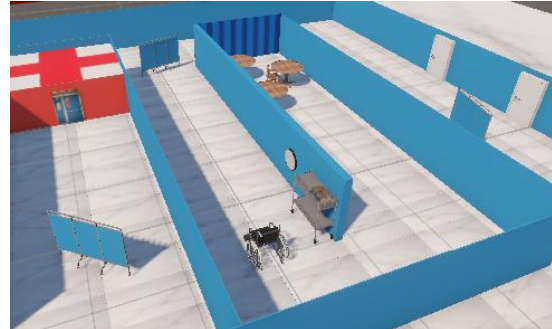
# Realistic Environments / Overall Results

## Can the project's solution approach reality?

For maximizing the realistic/dynamic component of the project's implementation, all the obstacles (medical props) in the scene are randomly & procedurally placed in the scene on launch, right before generating the grid and the unwalkable mask.



(run 1)



(run 2)

## Project's Outcome:

| What works                                          | Limitations                                     | Future Potential                                |
|-----------------------------------------------------|-------------------------------------------------|-------------------------------------------------|
| Agent reaches all three destinations reliably       | Moving obstacles during runtime isn't supported | Dynamic obstacles while running (not on launch) |
| Obstacle avoidance functions correctly              | No agent-to-agent collision (single agent only) | Multi-agent support, each with own hit box      |
| Stuck recovery resolves edge cases within 3 seconds | Simplified hospital layout                      | Full hospital layout                            |



Thank you for  
your attention.

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